

Module 12: Recovery with Concurrent Transactions and 2PC

AKTU BCS501 | Unit 5: Concurrency Control Techniques

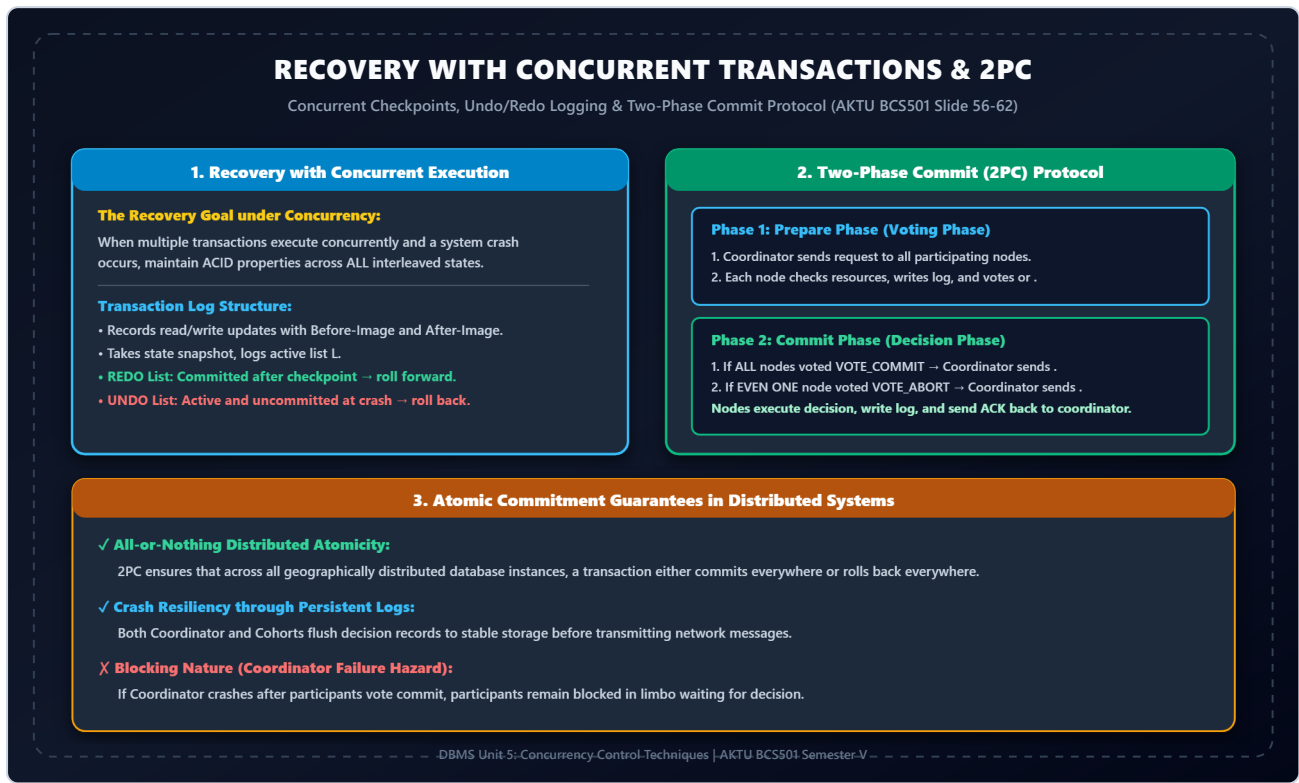


Figure 12.1: Concurrent Transaction Recovery, Checkpoints & 2PC Protocol

1. Concurrent Transaction Recovery

Maintains ACID properties during crashes. Active list L at checkpoint determines UNDO list (uncommitted active txns) and REDO list (committed txns after checkpoint).

2. Two-Phase Commit (2PC) Protocol

Phase 1 (Prepare): Coordinator sends PREPARE; nodes vote COMMIT or ABORT.

Phase 2 (Commit): If all vote yes, coordinator sends GLOBAL_COMMIT; else GLOBAL_ABORT.